10. **Many biological systems contain metal complexes. For example,**

(i) A red blood corpuscles (RBC) is composed of heme group, which is Fe^{2+}- Porphyrin complex.it plays an important role in carrying oxygen from lungs to tissues and carbon dioxide from tissues to lungs.

(ii) Chlorophyll, a green pigment present in green plants and algae, is a coordination complex containing Mg^{2+} as central metal ion surrounded by a modified Porphyrin ligand called corrin ring. It plays an important role in photosynthesis, by which plants converts CO_2 and water into carbohydrates and oxygen.

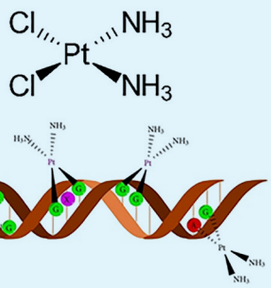
(iii) Vitamin B_{12}(cyanocobalamine) is the only vitamin consist of metal ion. it is a coordination complex in which the central metal ion is Co^{+} surrounded by Porphyrin like ligand.

(iv) Many enzymes are known to be metal complexes, they regulate biological processes. For example, Carboxypeptidase is a protease enzyme that hydrolytic enzyme important in digestion, contains a zinc ion coordinated to the protein.



Cisplatin:

Cisplatin is a square planar coordination complex (cis- $[Pt(NH_3)_2Cl_2]$), in which two similar ligands are in adjacent positions.



It is a Platinum-based anticancer drugThis drug undergoes hydrolysis and reacts with DNA to produce various crosslinks. These crosslinks hinder the DNA replication and transcription, which results in cell growth inhibition and ultimately cell death.

It also crosslinks with cellular proteins and inhibits mitosis.

Summary

- When two or more stable compounds in solution are mixed together and allowed to evaporate, in certain cases there is a possibility for the formation of double salts or coordination compounds. The double salts loose their identity and dissociates into their constituent simple ions in solutions , whereas the complex ion in coordination compound, does not loose its identity and never dissociate to give simple ions.
- According to Werner, most of the elements exhibit, two types of valence namely primary valence and secondary valence and each element tend to satisfy both the valences.In modern terminology, the primary valence is referred as the oxidation state of the metal atom and the secondary valence as the coordination number.
- Coordination entity is an ion or a neutral molecule, composed of a central atom, usually a metal and the array of other atoms or groups of atoms (ligands) that are attached to it.

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